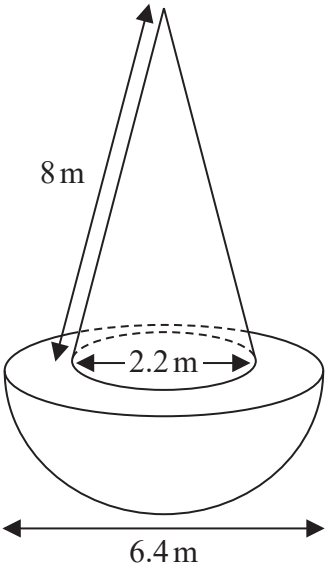
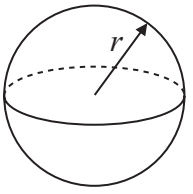
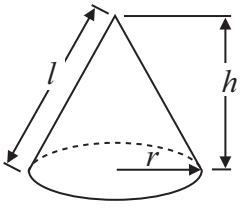
18 The centre of the base of a solid cone is placed at the centre of the flat face of a hemisphere to make a sculpture.



Surface area of sphere $= 4\pi r^2$



Curved surface area of cone $= \pi rl$



The diameter of the hemisphere is 6.4 m.
The diameter of the base of the cone is 2.2 m and the slant height of the cone is 8 m.

Callum is going to cover this sculpture with one coat of paint.
He assumes that each tin of paint will cover 8.5 m^2

Callum thinks he will need to buy 15 tins of paint.

(a) Will 15 tins of paint be enough?
You must show how you get your answer.

Hemisphere radius R = 3.2 m; cone radius r = 1.1 m.

Curved hemisphere = $2\pi R^2 = 20.48\pi$ m^2

1 mark
Half the curved sphere.

Curved cone = $\pi rl = \pi \times 1.1 \times 8 = 8.8\pi$ m^2

1 mark
Cone curved surface.

Exposed flat ring = $\pi(R^2 - r^2) = 9.03\pi$ m^2

1 mark
Include exposed flat ring.

Total area = $(20.48 + 8.8 + 9.03)\pi = 120.35441455...$ m^2

15 tins cover $15 \times 8.5 = 127.5$ m^2.

1 mark
Compare total coverage.

Yes. 127.5 > 120.3544..., so 15 tins are enough.

1 mark
Yes, with both areas.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Callum finds out that each tin of paint will actually cover 7.5 m^2

(b) How does this affect your answer to part (a)?

1 mark

More tins are needed.

15 tins now cover 112.5 m^2, less than 120.3544... m^2.

15 are not enough; 120.3544... ÷ 7.5 = 16.0472... → 17 tins.

(1)

(Total for Question 18 is 6 marks)